

MSC

HPMS Data Dictionary

Field-level definitions, types, constraints, and validation logic

v1.0 • 10 April 2026 • Author: Jubril

Legend: Required / Unique columns — Green = Yes, Grey = No, Amber = Conditional (see Notes column for logic).

1. Promoter Level Entity

Covers all three tiers: MSC Field Lead (Level 1), MSC Field MSC Field Agent (Level 2), MSC Frontline Rep (Level 3). Stored in a single promoters table with a level field to differentiate.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
promoter_id	UUID	Yes	Yes	System-generated UUID on account creation. Cannot be edited.	Primary key.
full_name	String	Yes	No	Min 2 chars, max 100 chars. Letters, spaces, hyphens only.	Display name across all dashboards.
phone_number	String	Yes	Yes	Must match Nigerian format: +234XXXXXXXXXX. Validated at input. No two promoters share a number.	Used for login and notifications.
email_address	String	No	Yes	Valid email format. Unique if provided. Optional for MSC Frontline Reps.	Required for MSC Field Leads; optional for MSC Field Agents and MSC Frontline Reps.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
level	Enum	Yes	No	Accepted values: FIELD_LEAD, FIELD_AGENT, FRONTLINE_REP. Set at creation; cannot be changed by the promoter.	Determines commission split logic and recruitment permissions.
parent_id	UUID/FK	Cond	No	Required for FIELD_AGENT (references a FIELD_LEAD) and FRONTLINE_REP (references an FIELD_AGENT). Null for FIELD_LEAD.	Foreign key to promoters table. Enforces hierarchy depth.
barcode_id	UUID/FK	Yes	Yes	Auto-assigned on account creation. Links to barcodes table.	Each promoter has exactly one barcode.
territory	String	Yes	No	Free text, max 100 chars. Represents assigned state or region.	Cross-region assignment requires Admin approval flag.
cross_region_approved	Boolean	Yes	No	Default: false. Set to true by Admin only when cross-region assignment is approved.	Checked before allowing activity outside primary territory.
status	Enum	Yes	No	Values: ACTIVE, SUSPENDED, INACTIVE, PENDING. Default: PENDING on creation.	SUSPENDED freezes all commission accrual immediately.
suspension_reason	Text	Cond	No	Required when status is set to SUSPENDED. Max 500 chars.	Stored for audit log. Admin-only field.
bank_account_number	String	No	No	10-digit Nigerian bank account number. Validated via NUBAN checksum.	Required before first commission payout can be processed.
bank_name	String	No	No	Free text from approved bank list dropdown.	Paired with account number for payout.
created_by	UUID/FK	Yes	No	References the Admin or Promoter who created this account.	FIELD_LEAD created_by = Admin ID. FIELD_AGENT = MSC Field Lead ID.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
					FRONTLINE_REP = MSC Field Agent ID.
created_at	Timestamp	Yes	No	System-generated UTC timestamp at record creation. Immutable.	
updated_at	Timestamp	Yes	No	Auto-updated on any field change.	

2. Barcode Entity

Every promoter is assigned one barcode. Sub-barcode for MSC Field Agents and MSC Frontline Reps are derived from the parent MSC Field Lead's barcode and stored here with a parent reference.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
barcode_id	UUID	Yes	Yes	System-generated UUID. Primary key.	
barcode_code	String	Yes	Yes	Unique alphanumeric string, system-generated. Format: MSC-[LEVEL_CODE]-[6-digit random]. e.g. MSC-CH-004821.	LEVEL_CODE: CH = MSC Field Lead, AG = MSC Field Agent, AM = MSC Frontline Rep.
qr_payload	String	Yes	Yes	Encoded string embedded in QR image. Contains barcode_code + checksum.	Used by scanner to validate authenticity.
promoter_id	UUID/FK	Yes	Yes	References the promoter this barcode belongs to. One-to-one.	Cannot be reassigned to another promoter.
parent_barcode_id	UUID/FK	Cond	No	Null for MSC Field Leads. Required for MSC Field Agents (references MSC Field Lead	Enables traceability up the chain.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
				barcode) and MSC Frontline Reps (references MSC Field Agent barcode).	
root_barcode_id	UUID/FK	Yes	No	Always references the MSC Field Lead barcode at the top of the chain. Same as barcode_id for MSC Field Leads.	Used for fast attribution queries without traversing tree.
status	Enum	Yes	No	Values: ACTIVE, REVOKED. Default: ACTIVE. Revoked barcodes cannot be scanned.	Barcode is revoked when promoter is permanently deactivated.
created_at	Timestamp	Yes	No	System-generated. Immutable.	

3. Pharmacy Entity

Duplicate detection logic: A pharmacy is considered a duplicate if ANY of the following already exist in the system: (1) business_registration_number, (2) exact match on pharmacy_name + street_address + lga. Both checks run on submission. If either match is found, registration is blocked and flagged for Admin review.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
pharmacy_id	UUID	Yes	Yes	System-generated UUID. Primary key.	
pharmacy_name	String	Yes	No	Min 3 chars, max 150 chars. Trimmed and stored in title case.	Not unique alone — combined with address for duplicate check.
business_reg_number	String	Yes	Yes	CAC registration number. Format: RC followed by 6–7 digits e.g. RC123456. Unique across all pharmacies.	PRIMARY duplicate detection field. Checked first.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
nafdac_licence_number	String	No	Yes	Format: NAFDAC/PHARM/XXXXX. Unique if provided.	Optional at registration; may be required before first order.
legal_rep_name	String	Yes	No	Full name of the pharmacy owner or legal representative. Min 2 chars, max 100 chars.	
legal_rep_phone	String	Yes	No	Nigerian format: +234XXXXXXXXXX. Not required to be unique (same person can manage multiple pharmacies).	Used for verification calls.
legal_rep_id_type	Enum	Yes	No	Values: NIN, BVN, PASSPORT, DRIVERS_LICENCE, VOTERS_CARD.	Used for KYC during verification.
legal_rep_id_number	String	Yes	No	Value of the selected ID. Validated by format per ID type. Not globally unique (shared reps possible).	NIN: 11 digits. BVN: 11 digits. Passport: 2 letters + 7 digits.
street_address	String	Yes	No	Full street address. Max 200 chars.	Combined with pharmacy_name + lga for SECONDARY duplicate check.
lga	String	Yes	No	Local Government Area. Selected from approved LGA list per state.	
state	String	Yes	No	Nigerian state. Selected from approved state list.	
onboarding_barcode_id	UUID/FK	Yes	No	References the barcode scanned during registration. Immutable after registration.	Determines commission attribution for all future transactions.
onboarding_promoter_id	UUID/FK	Yes	No	Resolved from onboarding_barcode_id at registration time. Stored directly for fast queries.	Immutable after registration.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
verification_status	Enum	Yes	No	Values: PENDING, VERIFIED, REJECTED. Default: PENDING.	Onboarding bonus is only triggered after status = VERIFIED.
verification_date	Timestamp	Cond	No	Set when verification_status changes to VERIFIED or REJECTED. Null while PENDING.	
verified_by	UUID/FK	Cond	No	References the Admin who verified. Required when verification_status = VERIFIED or REJECTED.	
rejection_reason	Text	Cond	No	Required when verification_status = REJECTED. Max 500 chars.	
first_order_settled_at	Timestamp	Cond	No	Set when the pharmacy's first order reaches SETTLED status. Null before that.	Triggers onboarding bonus calculation when set.
onboarding_bonus_paid	Boolean	Yes	No	Default: false. Set to true once the ₦2,000 bonus has been disbursed. Cannot revert to false.	Guard against double-payment.
registration_timestamp	Timestamp	Yes	No	UTC timestamp at the exact moment the barcode was scanned. Immutable.	Ownership timestamp — used to resolve any future attribution disputes.
created_at	Timestamp	Yes	No	System-generated at record creation.	
updated_at	Timestamp	Yes	No	Auto-updated on any field change.	

4. Order Entity

Only SETTLED orders contribute to commission calculations. Orders in any other status are excluded from monthly commission batch runs.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
order_id	UUID	Yes	Yes	System-generated UUID. Primary key.	
pharmacy_id	UUID/FK	Yes	No	References the pharmacy that placed the order.	
order_value	Decimal	Yes	No	Total monetary value of the order in NGN. Min: 0.01. Max: no cap.	2 decimal places. Used in commission calculation.
order_status	Enum	Yes	No	Values: PENDING, PROCESSING, SETTLED, FAILED, CANCELLED. Default: PENDING.	Only SETTLED contributes to commission.
settled_at	Timestamp	Cond	No	Set when order_status transitions to SETTLED. Null for other statuses.	Commission batch uses this field to determine the billing month.
settlement_ref	String	Cond	Yes	Payment gateway or internal reference for settlement. Required when status = SETTLED.	Audit trail for every commission-generating transaction.
billing_month	String	Cond	No	Format: YYYY-MM. Derived from settled_at. Set by system when order is settled.	Commission batch groups by this field.
created_at	Timestamp	Yes	No	System-generated. Immutable.	
updated_at	Timestamp	Yes	No	Auto-updated on status change.	

5. Commission Record Entity

One commission record is created per promoter per pharmacy per billing month. The batch engine creates these records; they are never created manually.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
commission_id	UUID	Yes	Yes	System-generated UUID. Primary key.	
promoter_id	UUID/FK	Yes	No	The promoter receiving this commission.	
pharmacy_id	UUID/FK	Yes	No	The pharmacy whose sales generated this commission.	
billing_month	String	Yes	No	Format: YYYY-MM. The month this commission covers.	Composite unique key: promoter_id + pharmacy_id + billing_month.
total_pharmacy_sales	Decimal	Yes	No	Sum of all SETTLED order values for this pharmacy in the billing month.	
commission_type	Enum	Yes	No	Values: DIRECT, OVERRIDE. DIRECT = promoter onboarded the pharmacy. OVERRIDE = earned from downstream level.	Determines split logic applied.
gross_commission	Decimal	Yes	No	Full commission before split: 1% of first ₦1M + 2% of amount above ₦1M.	
split_percentage	Decimal	Yes	No	Percentage of gross_commission this promoter earns. Values: 100, 90, or 10.	100 = direct MSC Field Lead. 90 = direct MSC Field Agent or MSC Frontline Rep. 10 = override.
net_commission	Decimal	Yes	No	$\text{gross_commission} \times (\text{split_percentage} / 100)$. The actual amount owed to this promoter.	
status	Enum	Yes	No	Values: CALCULATED, APPROVED, DISPUTED, PAID. Default: CALCULATED.	Batch sets to CALCULATED. Admin approves → APPROVED. Paid → PAID.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
dispute_note	Text	Cond	No	Required when status = DISPUTED. Max 500 chars.	
calculated_at	Timestamp	Yes	No	Timestamp of when the batch engine created this record.	
approved_by	UUID/FK	Cond	No	References Admin who approved. Required when status = APPROVED.	
approved_at	Timestamp	Cond	No	Set when status transitions to APPROVED.	

6. Onboarding Bonus Entity

One bonus record per pharmacy, ever. The record is created when the pharmacy is first verified and updated (not recreated) when the first order settles. Prevents double-payment by design.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
bonus_id	UUID	Yes	Yes	System-generated. Primary key.	
pharmacy_id	UUID/FK	Yes	Yes	One-to-one with pharmacy. A pharmacy can only ever have one bonus record.	Unique constraint enforces no double payment.
total_bonus_amount	Decimal	Yes	No	Always ₦2000.00. Stored explicitly for audit clarity.	
recipient_1_id	UUID/FK	Yes	No	The promoter who directly onboarded the pharmacy (MSC Frontline Rep, MSC Field Agent, or MSC Field Lead).	Always receives 90% = ₦1,800. If MSC Field Lead direct = 100% = ₦2,000.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
recipient_1_amount	Decimal	Yes	No	≠2,000 if MSC Field Lead direct. ≠1,800 if MSC Field Agent or MSC Frontline Rep.	
recipient_2_id	UUID/FK	Cond	No	The supervising promoter one level up. Null if MSC Field Lead onboarded directly.	Receives 10% = ₦200.
recipient_2_amount	Decimal	Cond	No	≠200 if applicable. Null if MSC Field Lead onboarded directly.	
trigger_status	Enum	Yes	No	Values: WAITING_VERIFICATION, WAITING_FIRST_ORDER, READY_TO_PAY, PAID. Default: WAITING_VERIFICATION.	System progresses this automatically.
verification_met_at	Timestamp	Cond	No	Set when pharmacy verification_status = VERIFIED.	
first_order_met_at	Timestamp	Cond	No	Set when pharmacy first_order_settled_at is populated.	Both conditions must be met before trigger_status = READY_TO_PAY.
paid_at	Timestamp	Cond	No	Set when bonus is disbursed. trigger_status must be READY_TO_PAY first.	
created_at	Timestamp	Yes	No	Set when pharmacy record is first created.	

7. Audit Log Entity

Immutable. No record in this table can be edited or deleted by any user, including Super Admins. Retained for a minimum of 12 months.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
log_id	UUID	Yes	Yes	System-generated. Primary key.	
actor_id	UUID/FK	Yes	No	The user who performed the action. References promoters or admins table.	
actor_role	Enum	Yes	No	Values: ADMIN, FIELD_LEAD, FIELD_AGENT, FRONTLINE_REP, SYSTEM.	SYSTEM used for batch and automated actions.
action_type	Enum	Yes	No	Values: ACCOUNT_CREATED, ACCOUNT_SUSPENDED, ACCOUNT_REINSTATED, PHARMACY_VERIFIED, PHARMACY_REJECTED, COMMISSION_APPROVED, BONUS_PAID, BARCODE_GENERATED, BARCODE_REVOKED, DISPUTE_RAISED, DISPUTE_RESOLVED.	Extensible list — new values require a schema migration.
entity_type	Enum	Yes	No	The type of record affected. Values: PROMOTER, PHARMACY, ORDER, COMMISSION, BONUS, BARCODE.	
entity_id	UUID	Yes	No	The ID of the specific record affected.	
previous_state	JSON	No	No	Snapshot of the entity's relevant fields before the action. Null for creation events.	Stored as JSON blob for flexibility.
new_state	JSON	No	No	Snapshot of the entity's relevant fields after the action. Null for deletion events.	
reason	Text	Cond	No	Required for: ACCOUNT_SUSPENDED, PHARMACY_REJECTED, DISPUTE_RAISED,	

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
				DISPUTE_RESOLVED. Max 500 chars.	
occurred_at	Timestamp	Yes	No	UTC timestamp. System-generated. Immutable.	The authoritative time of the action.
ip_address	String	No	No	IP address of the actor's session at the time of the action. Null for SYSTEM actions.	

8. Verification Call Log Entity

Tracks Admin-initiated verification calls to pharmacies. Each call is a separate log record. Multiple calls can exist per pharmacy. Used for compliance auditing and to identify pharmacies overdue for re-verification.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
call_log_id	UUID	Yes	Yes	System-generated. Primary key.	
pharmacy_id	UUID/FK	Yes	No	References the pharmacy being verified. Many call logs can exist per pharmacy.	Not unique — multiple calls allowed per pharmacy.
admin_id	UUID/FK	Yes	No	References the Admin who made the call.	
call_date	Date	Yes	No	Date the call was made. Format: YYYY-MM-DD. Cannot be a future date.	Used for filtering pharmacies overdue for re-verification.
call_outcome	Enum	Yes	No	Values: CONFIRMED, SUSPICIOUS, NO_ANSWER, WRONG_NUMBER, CALLBACK_REQUESTED. Cannot be null.	CONFIRMED = pharmacy verified as legitimate. SUSPICIOUS = flagged for further review.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
notes	Text	No	No	Admin's free-text notes from the call. Max 1000 chars.	Optional but encouraged for SUSPICIOUS and CALLBACK_REQUESTED outcomes.
follow_up_required	Boolean	Yes	No	Default: false. Set to true automatically when outcome = SUSPICIOUS or CALLBACK_REQUESTED.	Admin can also manually set to true for any outcome.
follow_up_date	Date	Cond	No	Required when follow_up_required = true. The date by which a follow-up call must be made. Format: YYYY-MM-DD.	
follow_up_done	Boolean	Yes	No	Default: false. Set to true when a subsequent call log is created for the same pharmacy after this one.	System auto-sets when next call log is created for this pharmacy.
created_at	Timestamp	Yes	No	System-generated at record creation. Immutable.	

9. Order Settlement Event Entity

A supplementary event log that records every status transition for an order. While the Order entity stores the current status, this table gives a full timeline of how an order moved from PENDING to SETTLED (or FAILED/CANCELLED). Used by the commission engine to confirm settlement integrity and by QA/Finance for dispute resolution.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
event_id	UUID	Yes	Yes	System-generated. Primary key.	

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
order_id	UUID/FK	Yes	No	References the order this event belongs to. Many events per order.	
pharmacy_id	UUID/FK	Yes	No	Denormalised from order for fast querying. References the pharmacy.	
from_status	Enum	Yes	No	The order status before this transition. Values: PENDING, PROCESSING, SETTLED, FAILED, CANCELLED. Null for first event (creation).	Null only on ORDER_CREATED event.
to_status	Enum	Yes	No	The order status after this transition. Same enum values as from_status.	SETTLED is the terminal state that triggers commission eligibility.
event_type	Enum	Yes	No	Values: ORDER_CREATED, PAYMENT_INITIATED, PAYMENT_CONFIRMED, ORDER_SETTLED, ORDER_FAILED, ORDER_CANCELLED.	Maps to the to_status transition.
settlement_ref	String	Cond	Yes	Payment gateway or internal reference. Required when event_type = ORDER_SETTLED. Unique across all settlement events.	Prevents duplicate settlement events for the same payment.
triggered_by	Enum	Yes	No	Values: SYSTEM, PAYMENT_GATEWAY, ADMIN. Identifies what caused the status transition.	ADMIN transitions are logged in audit_log as well.
gateway_response	JSON	No	No	Raw response payload from payment gateway. Null for non-payment transitions.	Stored for dispute resolution. Encrypted at rest.
commission_eligible	Boolean	Yes	No	Default: false. Set to true only when event_type =	Commission batch queries this field as the primary filter.

Field Name	Type	Required	Unique	Validation Rules	Notes / Logic
				ORDER_SETTLED AND order value > 0.	
billing_month	String	Cond	No	Format: YYYY-MM. Derived from occurred_at when commission_eligible = true. Null otherwise.	Used to group settled orders into commission billing periods.
occurred_at	Timestamp	Yes	No	UTC timestamp of the status transition. System-generated. Immutable.	The authoritative timestamp for commission billing month assignment.

End of Document — MSC HPMS Data Dictionary v1.0 (9 Entities)